

System Documentation

Harare Smart Urban Security and Monitoring System (HSUSMS)

Version 1.0 - Emerging Smart City Security Platform

1. Executive Summary

HSUSMS integrates CCTV surveillance, IoT sensors, AI analytics, emergency dispatch, traffic monitoring, facial recognition (demo), citizen mobile reporting, and OpenCV motion detection into a Central Command and Control Center for Harare, Zimbabwe.

2. System Aim

To design a centralized smart urban security system that enhances public safety, improves emergency response, supports law enforcement, and contributes to smart city development.

3. Architecture Layers

- 1) Data Collection: cameras, sensors, citizen app
- 2) Communication: REST API (FastAPI)
- 3) Processing: AI simulation, OpenCV vision, analytics
- 4) Command Center: web GIS dashboard
- 5) Response: police, ambulance, fire, disaster units

4. Security Zones

- Zone 1 - CBD: CCTV, facial recognition
- Zone 2 - Intersections: traffic cameras
- Zone 3 - Residential: patrol, parking sensors
- Zone 4 - Industrial: access control
- Zone 5 - Public Transport: crowd, noise monitoring
- Zone 6 - Government: biometric perimeter

5. Core Modules

Smart CCTV | IoT Sensors | Emergency Dispatch | Traffic Management | Facial Recognition (demo) | Citizen Reporting | OpenCV Vision | Crime Analytics

6. OpenCV Vision Module

Upload images (JPG, PNG) or videos (MP4, AVI) for real motion detection using OpenCV. The system detects motion regions, returns annotated frames, and optionally creates incidents in the command center database.

7. Citizen Mobile App (Flutter)

Features: emergency panic button, incident reporting with GPS, public alerts feed. Connects to the same REST API as the command center.

8. Functional Requirements

Real-time monitoring, suspicious activity detection, traffic congestion, emergency alerts, secure data storage, GIS mapping, mobile access, crime statistics, multi-agency coordination - all implemented in prototype.

9. Non-Functional Requirements

Scalable modular API, security via validation, 24/7 operation, privacy via anonymous reporting. Production: add JWT auth, PostgreSQL, encryption at rest, redundancy.

10. Technology Stack

Backend: Python, FastAPI, SQLAlchemy, SQLite, OpenCV, NumPy

Frontend: HTML/CSS/JS, Leaflet

Mobile: Flutter

Docs: Markdown, FPDF2

11. API Endpoints (Key)

/api/health | /api/dashboard/stats | /api/cameras | /api/sensors | /api/incidents | /api/dispatch/assign | /api/vision/analyze-image | /api/vision/analyze-video | /api/citizen/reports

12. Smart City Impact

KPIs: active incidents, response availability, sensor uptime, zone risk scores. Benefits: faster detection, evidence-based policing, citizen engagement. Risks: surveillance overreach (mitigate with governance), infrastructure gaps (phased rollout).

13. Implementation Phases

Phase 1: CBD pilot (current)

Phase 2: Intersections + IoT

Phase 3: Zones 4-6

Phase 4: Production RTSP, TensorFlow models, 5G edge

14. Conclusion

HSUSMS provides a complete academic and technical foundation for evaluating how urban security technologies contribute to Harare's development as an emerging smart city.

Appendix: Running the System

1. Start server: `.\run.ps1` from project root
2. Open dashboard: `http://127.0.0.1:8000`
3. API docs: `http://127.0.0.1:8000/docs`
4. Flutter app: `cd mobile\hsusms_citizen && flutter run`
5. Regenerate this PDF: `python docs\generate_pdf.py`